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Magnetostratigraphic constraints on two-stage eruptions of the Emeishan continental flood basalts

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[1] Temporal and spatial variations of high and low contents of Ti basalts in the Permian Emeishan large igneous province (ELIP) of southwest China and Vietnam are crucial for modeling of continental basalt formation and the nature of mantle-lithosphere interaction. A combined geochemical and magnetostratigraphic study was carried out at Yanyuan in the inner zone of the ELIP. Geochemical results indicate that the basalts are low-Ti (LT) basalts (Ti/Y ~ 484.2–511.4) in the Luomapu section and high-Ti (HT) basalts (Ti/Y ~ 558.2–658.9) in the Gongmushan section. Systematic paleomagnetic sampling and stepwise thermal demagnetization demonstrate that the higher temperature component shows two distinct characteristic remanent magnetizations with northeasterly upward and SWW downward directions. These results indicate a simple magnetostratigraphic normal and reversed polarity pattern from the lower and upper parts of the section, respectively. By comparing with available magnetostratigraphic and paleomagnetic data carried out in the ELIP, these results imply that the eruption of the Emeishan basalts can be divided into early (normal polarity subchron) and late (reversal subchron) stages. The early stage included LT basalts in the inner zone and most of the HT basalts in the intermediate and outer zones, and the late stage contained a part of LT and HT basalts in the upper parts of the sections in the inner zone and a few in the intermediate zone of the ELIP. Correlating with available Middle-Late Permian magnetostratigraphy and the Permian magnetic time scale, the new magnetic polarity sequence result suggests that the ELIP formed over a very short time interval (<1 Myr).

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1. Introduction

[2] The Middle to Upper Permian Emeishan continental flood basalts in southwestern China and northern Vietnam form a major part of the Emeishan large igneous province (ELIP) [e.g., *Lin*, 1985; *Zhang et al.*, 1988; *Chung and Jahn*, 1995; *Song et al.*, 2001a, 2001b; *Ali et al.*, 2005; *Xu et al.*, 2008; *Ukstins Peate and Bryan*, 2008], which covers an area of 0.5×10^6 km^2 [*Xiao et al.*, 2004] from the eastern margin of the Tibetan Plateau to the western margin of the Yangtze block. Geochemical studies reveal the characteristic temporal and spatial distributions of high-Ti (HT) and low-Ti (LT) basalts in the ELIP, where the dominant low-Ti basalts with low Ti/Y (<500) and εNd(t) (−0.34 ~ −3.76) erupted earlier than the high-Ti basalts with high Ti/Y (>500) and εNd(t) (−1.17 ~ 0.43), as they occur in the lower section in the western part of the ELIP, whereas the subordinate high-Ti basalts are distributed in both the upper sections of the western part, and uniformly in the eastern part of the ELIP [*Xu et al.*, 2001; *Xiao et al.*, 2004]. It has been argued that both varieties originated from a mantle plume, with more extensive crustal contamination of the low-Ti than in the high-Ti basalts. The distribution of the high- and low-Ti basalts coincides with the recently proposed lithospheric domed area postulated to have been formed by a mantle plume, where the western and eastern part of the ELIP correspond to the inner and intermediate to outer zones of the plume, respectively [*He et al.*, 2003, 2007]. However, the proposition of a regional-scale up-doming structure was questioned by *Ukstins Peate and Bryan* [2008]. Recently published radiometric ages for the ELIP suggest an age range of eruptions from 263 to 257 Ma with error estimates of 4 Myr [*He et al.*, 2007; *Xu et al.*, 2008].

[3] *Y. G. Xu et al.* [2007] argued that the thick low-Ti lavas in the inner zone may record the early main episode of flood basalt emplacement, whereas the overlying thin high-Ti basalts represent the waning activity of the plume and thus were products of low degrees of mantle melting beneath relatively thicker lithosphere with a lower geotherm. *Maury et al.* [2003] also suggested that the high- and low-Ti basalts might have originated from distinct mantle sources under variable melting conditions, e.g., high-Ti basalts resulted from low melting degrees of the enriched rim of a mantle plume that were emplaced in a proximal position with respect to the continent, and low-Ti basalts originated closer to the plume core, either from higher degrees of melting or from a more depleted source, which were emplaced oceanward with respect to the high-Ti basalts. The temporal relationship between the low-Ti basalts of the inner zone and high-Ti basalts of the intermediate-outer zones of the ELIP remains unclear.

[4] The ELIP basalts were considered to be good paleomagnetic research objects because of stable remanence acquired by cooling from above the Curie temperature (Tc) in the presence of a magnetic field [e.g., *McElhinny et al.*, 1981; *Zhao and Coe*, 1987]. The Middle Permian was characterized by predominant reversed geomagnetic polarity alternating with normal ones [*Heller et al.*, 1995; *Ogg et al.*, 2008]. The eruptive duration of the ELIP may be constrained by the radiometric dating and magnetostratigraphy of the basalts [e.g., *Courtillot et al.*, 1986], the available magnetostratigraphic studies of which have been concentrated in the eastern margin of the intermediate zone [*Huang and Opdyke*, 1998; *Ali et al.*, 2002]. Because of the uncertainty of the Permian magnetic time scale, the magnetostratigraphic pattern (N-R) of the ELIP observed at Duge (Figure 1a) in western Guizhou suggested that the emplacement of the ELIP was less than two million years [*Huang and Opdyke*, 1998]. However, temporal relationships, petrological correlation of LT and HT basalts in the ELIP and whether HT basalts in the inner zone are equivalent to HT in the intermediate zone remain unconstrained, answers to which would be very important for modeling the formation and thermal structure of the ELIP and the nature of the mantle-lithosphere interaction.

[5] In this paper we first report a combined geochemical and magnetostratigraphic study at Yanyuan in the inner zone of the ELIP, and then, by comparison with previously published data, attempt to constrain the temporal relationship between the high-Ti

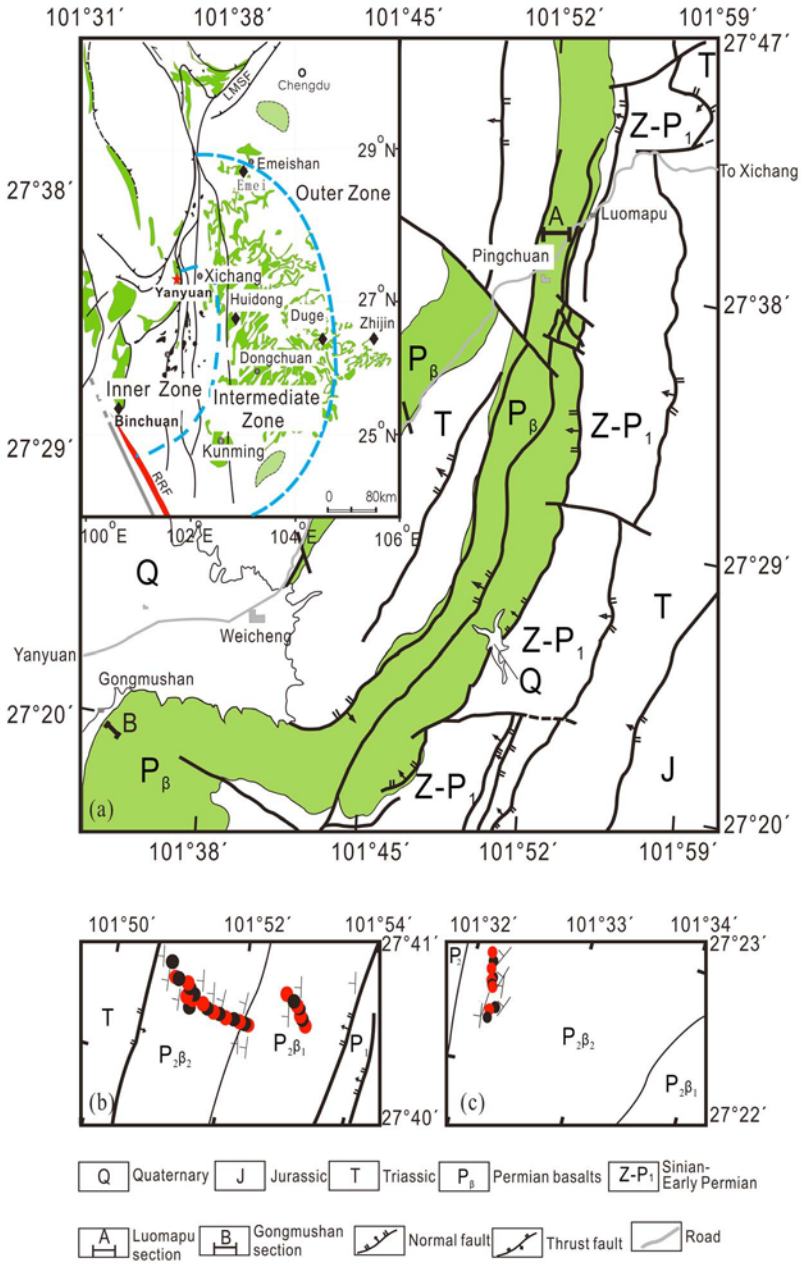


Figure 1. (a) Simplified geological map of the Yanyuan area of Sichuan province, southwestern China [*SBGMR*, 1990]. Green areas are Permian basalts. Inset illustrates the distribution of the Late Permian volcanic successions of the Emeishan large igneous province in southwestern China (modified from *Song et al.* [2008], copyright 2008, with permission from Elsevier). Abbreviations are as follows: RRF, Red River fault; LMSF, Longmenshan Fault. Dashed curves separate inner, intermediate, and outer zones of the ELIP [after *He et al.*, 2003]. Red star shows the sampling areas at Yanyuan, and diamonds denote the areas where paleomagnetic results were obtained. The detailed distribution and dip directions of the sampling sites in the (b) Luomapu and (c) Gongmushan sections.



Figure 2. Outcrop photographs of sampling section in Yanyuan indicate where the bedding directions were obtained.

and low-Ti basalts of the inner and intermediate/outer zones and the duration of the ELIP.

2. Geological Setting

[6] The Yanyuan and Binchuan areas are all situated in the inner zone of the ELIP on the western margin of the Yangtze block (Figure 1a). The lava succession in this area is thicker than those in the intermediate-outer zone. The basalts at Yanyuan are divided into two members: a lower member mainly composed of hyaloclastite and pyroclastic lava and upward to a gray-green massive basalt, amygdaloidal and porphyritic basalts; the upper member consists mainly of massive, amygdaloidal and porphyritic basalts, intercalated by thin sandstone and conglomerate in the uppermost part [*Sichuan Bureau of Geology and Mineral Resources* (*SBGMR*), 1990]. Bedding directions of the basalts were inferred by the dip directions of interbedded lava bed or thin sandstone (Figure 2). Paleomagnetic samples were collected using a portable core drill and oriented with a magnetic compass along two sections at Yanyuan.

[7] The Luomapu section (LM) (27°40.3′N, 101°52.5′E) is located along a dell near Luomapu village on the road from Xichang to Yanyuan (Figure 1b). The section shows the main succession of Emeishan flood basalts overlying the Lower-Middle Permian Maokou Limestone, which in turn is underlain by Lower Triassic clastic sediments. As noted above, the lower member of the section consists of amygdaloidal and/or vesicular basalts and massive basalts, upward to porphyritic basalts interbedded with thin purple-red sandstone in the lower part of upper member. The upper member is composed of porphyritic basalts and a very few interbedded purple-red sandstones, ending up with massive basalts. A series of columnar joints are visible on the top of the upper member. The faults are developed in the sampling area forming the contacts between the lowest basalts and the underlying limestone, and the uppermost basalts and overlying strata. A total of 345 relatively fresh samples (31 sites, each site includes usually more than 10 samples) were collected systematically from the base to the top of the section with an average interval about 3.8 m, with one large gap of about 230 m in the middle part where covering vegetation prevented sampling. Five gray (Lower Triassic) sandstone samples from the uppermost part of the section were also collected; five hand block samples from the section were sampled for geochemical study.

[8] The basalts in the Gongmushan section (GM) (27°22.3′N, 101°32.2′E) are overlain by Upper Permian sediments. These basalts were selected in order to perform a fold test and to complete the upper

part of the upper member that may be missing in the Luomapu section because of the thrusting fault (Figure 1c). The sequence is characterized by a predominance of massive basalt and then amygdaloidal and porphyritic basalts, which are overlain by clast-supported breccia. In this section, 118 samples (from 12 sites) covering 240 m in thickness were collected for magnetostratigraphic study and four block samples for geochemical work.

3. Geochemical Analysis and Results

[9] Samples were powdered in agate mortars to minimize potential contamination. Major elements were determined using an ARL9800XP X-ray fluorescence spectrometer (XRF) at the analytical center of Nanjing University. Trace elements, including REE, were analyzed using Inductively Coupled Plasma Mass Spectrometry (ICP-MS) at the State Key Laboratory for Mineral Deposits Research of Nanjing University. Analytical uncertainties are 1%–2% for major elements and 5% for trace elements. Details of the analytical procedures are given by *Gao et al.* [2003]. The major and trace element results are listed in Table 1.

[10] The basalt samples are variably altered with LOI below 3% in the LM and GM sections. The samples in section LM are LT basalts with TiO_2 (1.64–2.65 wt %) and Ti/Y ratios in the range of 484.2–511.4, which is consistent with data reported by *Xu et al.* [2004] for the Pingchuan section located just 6 km to the south. However, the series in section GM with TiO_2 (3.01 wt %–3.76 wt %) and Ti/Y = 558.2–658.9 are HT basalts. The major element composition is characterized by SiO_2 around 45.31 wt %–50.89 wt % and Fe_2O_3 (total) around 13.22 wt %–17.28 wt % (on average 15.37 wt %). The MgO contents are slightly different between the LM (MgO, 4.75 wt %–8.00 wt %; Mg #, 36.3–54.7) (Mg # = $Mg/(Mg+Fe^{2+})$) and GM sections (MgO, 3.92 wt %–5.25 wt %, Mg #, 31.9–40.4), indicating that the upper part of the series (represented in GM section) was more evolved than the lower part (LM section) (Table 1).

[11] In trace element compositions, all samples (N = 9) display average incompatible trace element ratios similar to ocean island basalts (OIB) [*Weaver*, 1991], including Zr/Nb = 5.26, La/Nb = 0.57, Ba/Nb = 7.08, Ba/Th = 83.76, Rb/Nb = 0.38, Th/Nb = 0.09, Th/La = 0.15, and Ba/La = 12.56. Basalt data from LM and GM sections show primitive mantle normalized [after *Sun and McDonough*, 1989] multi-element abundance similar to those observed in OIB basalts (Figure 3a), but differing from Mid-Ocean Ridge Basalt (MORB). Comparing with OIB, the basalts from both study sections show negative Sr anomalies, which are more distinct at GM than at LM. The basalts in both sections also show positive Nb anomalies but those at LM display stronger anomalies. The occurrence of negative Sr anomalies may be related to plagioclase fractionation.

[12] REE patterns are shown in Figure 3b. The HT basalts at section GM have higher REE contents (∑REE = 200.19–261.45) and stronger differentiation ($(La/Yb)_N$ = 8.54–10.41) than those of the LT basalts in section LM; the latter display a flatter pattern and lower light REE contents (∑REE = 69.58–126.05) ($(La/Yb)_N$ = 3.24–5.08).

4. Rock Magnetic and Paleomagnetic Results

[13] All samples were cut in the laboratory into a standard cylinder 2.3 cm long with a diameter of 2.54 cm. Progressive thermal demagnetization was employed for magnetic cleaning using a TD-48 thermal demagnetizer. Magnetization of the samples was measured in a JR6a and partial samples were measured in a 2G cryogenic magnetometer. All instruments are housed in a magnetically shielded room (<500 nT) in the Department of Earth Sciences at Nanjing University, which greatly decreases the effect of the present geomagnetic field during measurements. The thermal demagnetization temperature interval was 40–60°C at low temperature, which lowered at high temperature, to about 10–20°C. Vector endpoint diagrams [*Zijderveld*, 1967] were used to analyze the data for each sample and principal component analysis [*Kirschvink*, 1980] was used to determine characteristic remanent magnetization (ChRM) directions. The site mean directions were calculated either using the statistics of *Fisher* [1953] or by using the intersection of sector-constrained great circles combined with the directly observed HTC directions of the statistical method developed by *McFadden and McElhinny* [1988].

4.1. Rock Magnetic Results

[14] Temperature dependences of magnetic susceptibility experiments were conducted in order to test magnetic mineral transformations during laboratory heating. Susceptibility measurements were performed using a KLY3 Kappa bridge with a CS-3 high-temperature furnace in consecutive heating-cooling cycles with increasing maximum heating

Table 1. Major and Trace Element Data for Emeishan Basalts of the Luomapu and Gongmushan Sections, Yanyuan Area, China

	Sample								
	LM1	LM2	LM3	LM4	LM5	GM1	GM2	GM3	GM4
					Major Element (wt %)				
SiO_2	49.94	46.88	46.32	45.97	46.12	47.28	51.03	46.94	46.89
TiO_2	2.00	1.65	2.55	2.59	2.66	3.40	3.02	3.45	3.77
Al_2O_3	14.01	15.08	14.27	14.82	14.26	13.57	14.04	13.87	13.26
Fe_2O_3	13.56	13.31	16.12	17.29	16.73	15.54	13.53	15.93	16.74
MnO	0.19	0.34	0.22	0.28	0.23	0.17	0.17	0.21	0.20
MgO	5.95	8.05	5.32	5.23	4,77	5.25	4.05	5.22	3.92
CaO	7.62	8.10	10.72	6.21	8.34	6.72	6.17	9.86	8.39
Na_2O	3.86	2.91	2.91	4.02	3.99	4.33	4.97	2.64	2.77
K_2O	0.31	0.21	0.37	0.09	0.73	0.57	0.81	0.91	2.23
P_2O_5	0.18	0.14	0.37	0.27	0.30	0.35	0.34	0.40	0.48
LOI	2.89	3.96	1.32	3.26	2.30	2.78	2.14	0.81	1.44
Mg #	46.7	54.7	39.7	37.7	36.3	40.4	37.4	39.6	31.9
					Trace Element (ppm)				
Li	11.4	26.5	10.6	12.6	10.7	13.6	7.7	7.4	7.0
Be	3.1	2.6	6.3	5.1	5.0	9.1	8.7	8.1	11.9
Sc	25.7	28.0	22.7	25.6	24.2	20.3	18.2	19.8	19.6
Ti	11543	9588	15153	14953	15532	20016	17574	20264	22606
V	360	339	398	457	428	353	331	429	413
Cr	77.69	115.05	40.18	40.90	10.66	51.09	46.93	37.43	14.40
Mn	1328	2243	1557	1927	1630	1190	1194	1472	1420
Co	48	53	51	57	50	51	35	49	46
Ni	73	102	54	59	30	71	52	62	38
Cu	174	356	75	43	70	246	122	257	179
Zn	102	92	131	148	135	147	133	152	175
Ga	16.1	15.4	20.5	23.3	21.3	24.6	22.9	24.3	27.1
Rb	9.19	6.53	5.96	1.98	17.41	9.03	11.30	13.07	82.56
Sr	237	162	257	100	320	295	262	510	449
Y	23.22	19.66	28.26	30.88	30.37	32.85	31.48	30.75	39.76
Zr	134	95	147	137	165	275	237	232	344
Nb	16.9	15.7	31.7	25.8	37.7	52.9	46.5	54.1	75.8
Mo	8.89	5.78	3.30	1.84	2.06	2.09	1.94	1.84	2.35
In	0.08	0.07	0.08	0.09	0.09	0.09	0.08	0.10	0.12
Sn	1.54	1.13	1.75	1.67	1.86	2.47	2.37	2.11	3.03
Cs	0.58	1.26	0.23	0.17	0.69	0.02	0.07	0.15	2.30
Ba	210	114	240	68	317	222	312	395	546
La	11.1	8.5	14.5	13.9	19.2	37.2	32.3	30.0	39.2
Ce	35.4	26.2	37.4	38.0	46.5	101.6	92.7	87.9	116.5
Pr	4.03	2.98	4.84	4.62	5.55	10.85	9.88	9.39	12.39
Nd	18.8	13.9	21.9	21.4	25.5	45.6	42.2	39.9	51.5
Sm	4.51	3.38	5.19	5.25	5.96	8.60	8.06	7.67	9.85
Eu	1.41	1.18	1.74	1.75	2.01	2.60	2.49	2.65	3.06
Gd	4.76	3.75	5.64	5.84	6.31	8.30	7.66	7.57	9.56
Tb	0.77	0.64	0.93	0.95	1.04	1.18	1.12	1.13	1.43
Dy	4.81	3.95	5.78	5.97	6.27	7.02	6.41	6.65	8.44
Ho	0.83	0.71	1.03	1.06	1.11	1.12	1.05	1.09	1.40
Er	2.42	2.04	2.99	3.09	3.16	3.21	2.92	3.04	4.01
Tm	0.35	0.30	0.44	0.44	0.45	0.46	0.43	0.45	0.59
Yb	1.92	1.78	2.48	2.51	2.56	2.41	2.20	2.36	3.10
Lu	0.27	0.25	0.37	0.36	0.37	0.34	0.32	0.35	0.45
Hf	3.69	2.75	3.95	3.94	4.33	6.95	6.26	5.87	8.24
Ta	0.57	0.52	0.94	0.80	1.16	1.70	1.49	1.71	2.34
W	0.49	0.55	0.33	0.34	0.29	0.42	0.35	0.70	0.88
Pb	1.48	1.65	2.85	2.87	3.81	7.18	5.80	4.09	6.25
Bi	0.10	0.06	0.16	0.04	0.13	0.10	0.06	0.06	0.11
Th	1.60	1.33	1.83	1.61	2.76	6.26	5.56	4.80	7.43
U	0.37	0.25	0.37	0.24	0.25	1.21	1.12	1.06	1.61

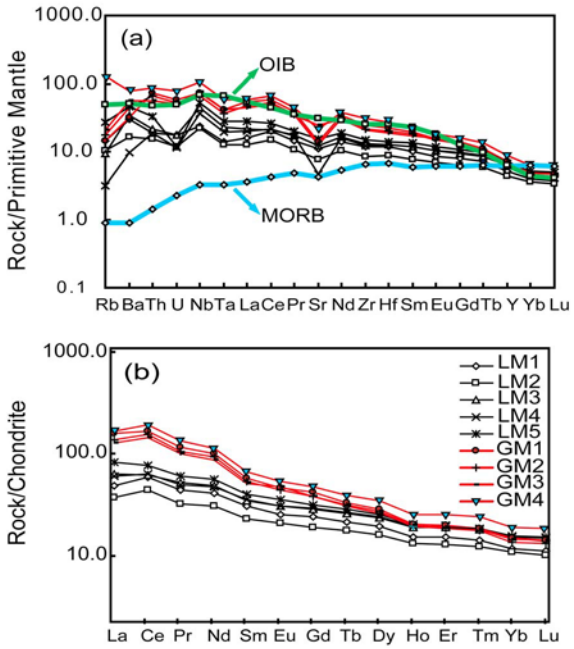


Figure 3. (a) Trace element and (b) REE diagrams of the Emeishan basalts at Yanyuan, Luomapu (LM) and Gongmushan (GM) sections. Normalizing data are from *Sun and McDonough* [1989].

temperatures up to 700°C in air (Figure 4). The $k-T$ curves show minor differences between the lower and upper part basalts in section LM. The selected samples from the lower part of the section display a progressive susceptibility drop during heating near 585°C (Figures 4a–4e), suggesting the existence of magnetite in these samples. The resultant irreversible curves indicate that the transformation of new magnetite occurred through an increase of susceptibility between 600°C and 500°C during the cooling process. This behavior indicates that magnetite is the major contributor to the susceptibility. High susceptibilities are obtained from the upper section (Figures 4f–4h). The $k-T$ curves of the basalts from the upper section exhibit irreversible changes at around 620°C, indicating a main carrier of titanohemite; in some oceanic basalts, a transformation hump of magnetic susceptibility between ~300°C and ~450°C is generally interpreted as the presence of (titano)maghemite formed during ionic reordering associated with low-temperature oxidation, which might carry a self-reversed component of natural remanent magnetization above room temperature [*Doubrovine and Tarduno*, 2004, 2006]. However, the $k-T$ experiments suggest that little titanomahemite is present in the basalts from the LM section.

[15] Magnetic hysteresis curves were measured using a MicroMag 2900 at the Department of Physics at Peking University; ten chips of representative samples were chose for hysteresis loop measurement. The saturation magnetization of ferromagnetic component (Ms), saturation remanent magnetization (Mrs), and coercivity (Bc) were determined after correcting high-field paramagnetic contribution. The coercivity of remanence (Bcr) was calculated from backfield demagnetization of the saturation remanence. The basalt samples from the lower and upper parts exhibit different hysteresis behavior, as shown in Figure 5. The hysteresis loops of the lower part basalts closed before 150 mT, and the basalts of the upper part of the section closed at 250 mT, indicating that low-coercivity minerals are the main carrier. The values of Mrs/Ms versus Bcr/Bc data show that the bulk magnetic properties of the basalt samples in the upper part of the section range from single domain (SD) to pseudo-single domains (PSD), with the basalts of the lower section covering from pseudo-single-domain (PSD) to multidomain (MD) behavior [*Day et al.*, 1977; *Dunlop*, 2002] (Figure 6). Hysteresis results generally follow the SD-MD mixing trends of *Dunlop* [2002].

4.2. Remanent Magnetic Analysis

[16] Two kinds of thermal demagnetization results were found in our basalt samples: one has a single component in some samples for which unblocking temperatures were up to 695°C, in which the magnetic direction is close to that of the present geomagnetic field; another has two components that are usually displayed after thermal demagnetizing in most samples. In the latter case, a first lower-temperature component (LTC) is isolated below 350°C from most samples. The mean direction is Dg = 0.1°, Ig = 42.4°, α_{95} = 1.7° before tilt correction, which closes to the present Earth field (PEF) direction (D = −1.3°, I = 41.6°, dipole field I = 46.3°). A second higher-temperature component (HTC) is determined between temperatures of 400°C and 590°C, and/or is unblocked even up to 695°C in some samples (about 12%). The thermal demagnetization of NRM in most samples shows a broad range of unblocking temperatures around 580°C, suggesting that the magnetic carrier is magnetite with a minor content of Ti. Otherwise, the unblocking temperature closes to 695°C, indicating the presence of hematite [*Lowrie*, 1990]. In some cases, the transformation of some magnetic minerals occurred during the heating process or the intensity is decreased down to the

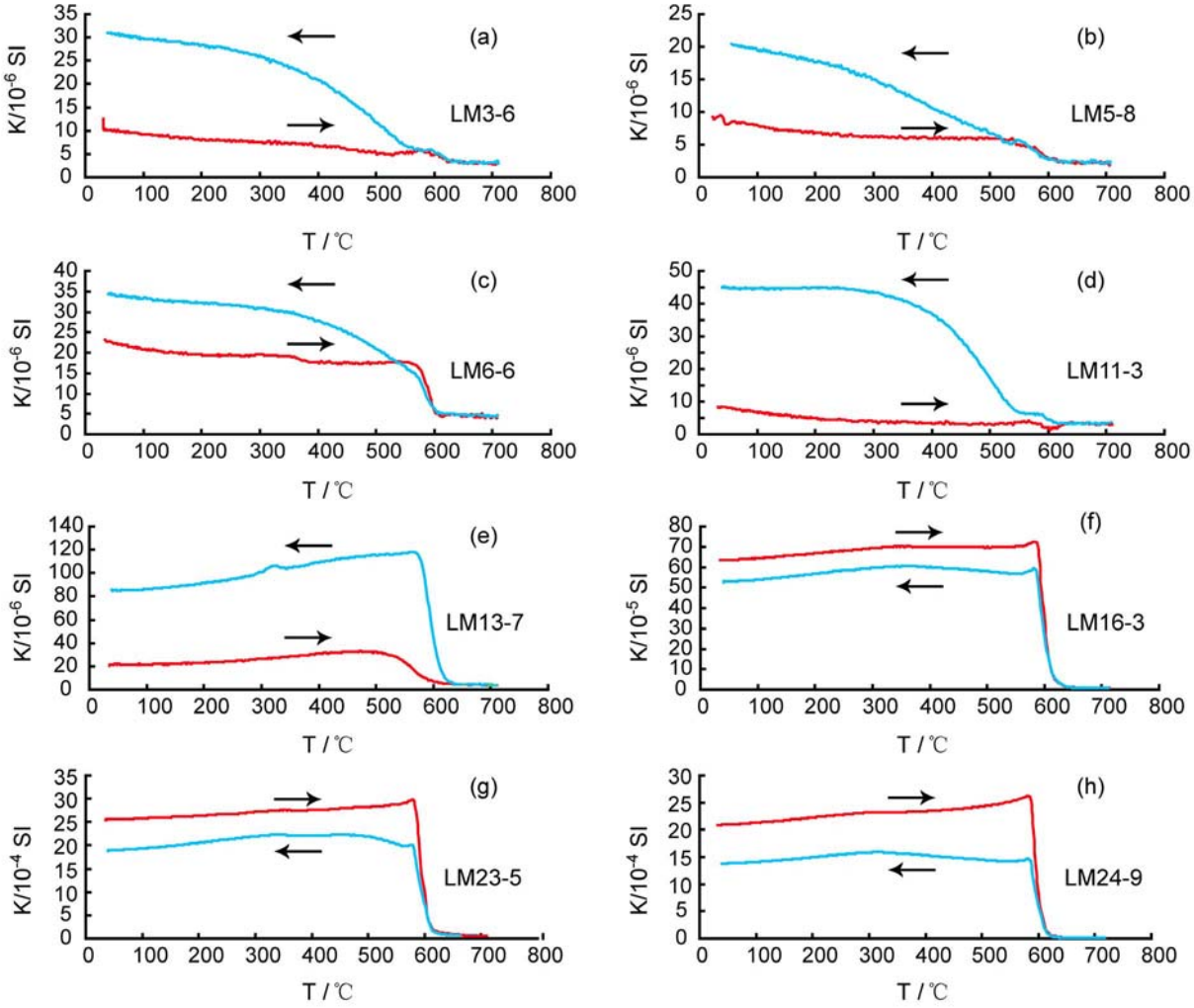


Figure 4. The $k - T$ curves of representative samples from the LM section. The curves were obtained in the atmosphere. (a–c) From the lower part and (d–h) from the upper part in the LM section. Sample name and site number are shown on each plot.

limit of the instrument, the higher-temperature component cannot be isolated. However, some of these samples show the well-defined great circle paths (Figures 7g–7j) that can be used for further statistic analysis [*McFadden and McElhinny*, 1988; *Heller et al.*, 1995].

4.3. Magnetic Results and Polarity Sequence

[17] The HTC shows two distinct directions and/or direction changes along the great circles in the LM section. One is a northeasterly direction with a shallow, negative inclination in *geographic* coordinate (Figures 7a–7c); 82 samples from 8 sites resolved the HTC and/or great circles. The site mean directions are Dg = −0.3°, Ig = −32.2°, k = 26.7, α_{95} = 10.9° before tilt correction, and Ds = 44.0°, Is = −18.1°, k = 26.7, α_{95} = 10.9° after tilt correction (Table 2). Another direction shows a nearly antipodal direction to the previous one, which is directed downward to the south in in situ coordinate (Figures 7d–7j). The site mean direction from 19 sites (168 samples) is Dg = 185.2°, Ig = 54.3°, k = 23.6, α_{95} = 7.1° before tilt correction, and Ds = 250.1°, Is = 23.7°, k = 27.2, α_{95} = 6.5° after tilt correction. The results from five sandstones were abandoned because the directions became erratic at higher temperatures.

[18] The HTCs show a good consistency from the GM section. The stable ChRM components were isolated from 12 sites (101 samples). The site mean direction is Dg = 268.9°, Ig = 42.3°, k = 57.3, α_{95} = 5.8° before tilt correction, and Ds = 279.3°, Is = 14.9°, k = 74.2, α_{95} = 5.1° after tilt correction (Table 2).

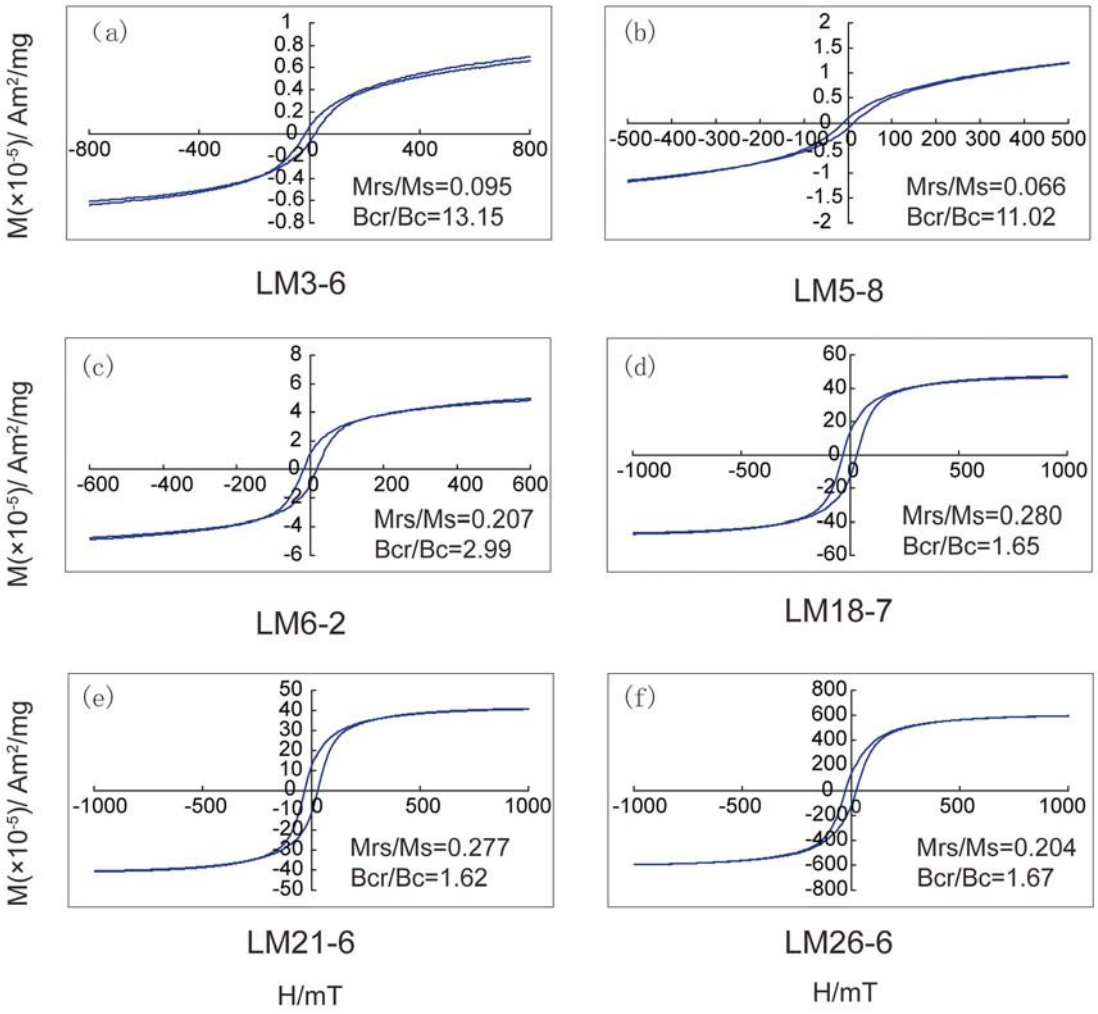


Figure 5. Examples of hysteresis loops from the lower and upper parts of the LM section. (a–c) Results from the lower part and (d–f) results from the upper part of the section. Abbreviations are as follows: Bcr, coercivity of remanence; Bc, coercivity; Mrs, saturation remanent magnetization; Ms, saturation magnetization.

[19] The fold test could not be effected in the LM section because the basalt flows have a monoclinal structure. However, a fold test was performed while the series at GM are combined and was positive at the 95% confidence level [*McFadden*, 1990], with $\xi_{2(\text{in situ})} = 27.25$, $\xi_{2(\text{tilt corrected})} = 5.4$, $\xi_c = 7.264$ (N = 39 sites for the both sections). In the other case, reversal directions obtained from both the LM and GM sections could also be used for performing a fold test, which also passed at the 99% confidence level with $\xi_{2(\text{in situ})} = 25.27$, $\xi_{2(\text{tilt corrected})} = 6.918$, $\xi_c = 9.157$ (N = 31 sites). All these results suggest that the HTCs were formed before fold deformation in the Triassic, implying primary remanences.

[20] The magnetic polarity sequence in the LM section consists simply of two magnetozones (Figure 8): a normal subchron zone of about 350 m

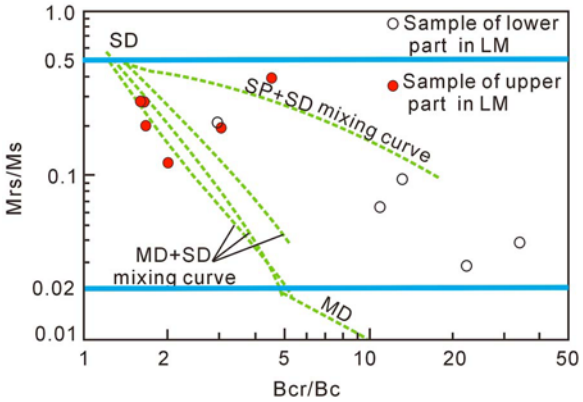


Figure 6. Magnetic hysteresis parameters plotted on a Day diagram [*Day et al.*, 1977; *Dunlop*, 2002]. Open (solid) symbols are results from the lower (upper) part of the LM section. SD, single domain; SP, superparamagnetic; MD, multidomain.

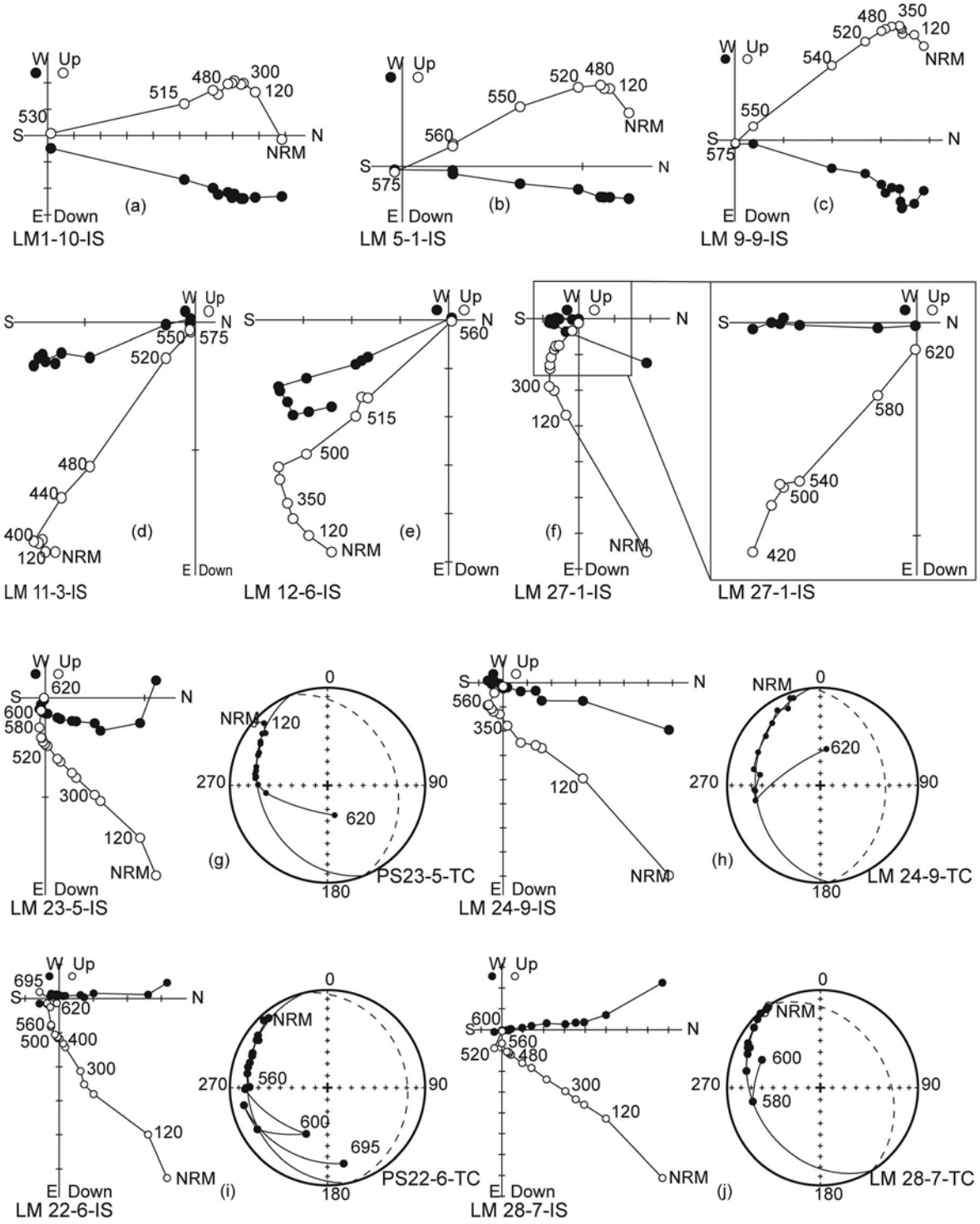


Figure 7. Representative orthogonal plots of thermal demagnetization results [*Zijderveld*, 1967] of basalt samples from Yanyuan in in situ coordinates. Plotted points represent the endpoints of the magnetization vector during progressive thermal demagnetization. Open (solid) symbols represent projections onto the vertical (horizontal) plane. Treatment levels are in °C. (a–c) Typical samples designed as normal polarity, (d–f) samples with reversed polarity, and (g–j) samples with heavy recent overprint in the orthogonal plots and stereoplot demonstrated magnetic direction changes along the great circles in stratigraphic coordinates.

Table 2. Site Mean Directions of the HTC Results From the Luomapu and Gongmushan Sections[a]

Site	Strike (deg)	Dip (deg)	n(g/d)/N	In Site Dec. (deg)	In Site Inc. (deg)	Tilt Corrected Dec. (deg)	Tilt Corrected Inc. (deg)	k	α_{95}	Polarity
Luomapu Section										
LM1–2*	193.7	78	15(2/13)/20	12.2	−16.5	29.6	−4.8	13.7	10.7	N
LM3–4*	193.7	78	13(1/12)/20	333.7	−20.8	31.0	−41.4	18.9	9.8	N
LM5	193.7	78	10(0/10)/10	8.9	−32.7	45.2	−10.4	51.5	6.8	N
LM6	193.7	78	8(0/8)/10	4.7	−43.6	56.0	−14.7	38.1	9.2	N
LM7	193.7	78	7(2/5)/10	1.5	−31.2	43.0	−16.5	27.4	11.7	N
LM8	193.7	78	10(2/8)/10	353.3	−42.8	55.3	−23.0	28.2	9.3	N
LM9	193.7	78	9(7/2)/10	351.1	−43.0	55.7	−24.6	8.4	19.7	N
LM10	193.7	78	10(2/8)/10	10.4	−22.3	35.0	−7.5	10.4	15.9	N
LM11	175.7	68	10(1/9)/10	164.8	55.1	227.7	24.0	18.5	12.3	R
LM12	212.7	66	9(1/9)/10	193.3	44.0	251.0	30.0	10.7	16.6	R
LM13	181.7	68	10(1/10)/11	180.7	51.2	230.6	17.6	87.1	5.3	R
LM14	205.7	71	8(0/8)/10	182.7	36.7	237.8	29.4	22.4	12.0	R
LM15	205.7	71	7(1/6)/10	200.8	37.5	240.7	15.2	16.6	17.2	R
LM16	215.7	62	8(4/4)/10	192.1	24.8	229.2	31.1	9.4	19.7	R
LM17	206.7	78	5(0/5)/8	183.4	46.0	252.1	24.7	21.9	16.7	R
LM18	186.7	51	8(3/5)/10	176.5	78.2	261.4	40.1	23.1	12.1	R
LM19	199.7	64	7(4/3)/11	181.4	61.4	257.5	31.1	23.4	13.6	R
LM20–21*	192.7	76	12(6/6)/20	186.8	51.1	242.5	14.3	6.0	17.6	R
LM22	197.7	68	6(4/2)/10	179.4	76.2	273.2	25.6	22.9	15.9	R
LM23–24*	194.7	72.1	17(12/5)/22	213.1	72.4	267.7	11.7	7.1	14.4	R
LM25	206.7	76	6(3/3)/10	185.8	30.3	233.9	24.9	127.4	6.4	R
LM26	192.7	78	8(4/4)/10	176.0	67.8	260.4	17.4	13.1	16.7	R
LM27	204.7	79	8(1/7)/10	186.0	42.6	246.3	21.1	75.6	6.5	R
LM28	190.7	69	7(4/3)/10	184.4	56.9	245.0	20.2	58.8	8.5	R
LM29	202.7	85	10(0/10)/11	198.2	68.2	270.9	6.3	7.3	19.5	R
LM30	202.7	76	9(6/3)/13	178.9	54.4	256.7	25.1	21.1	12.1	R
LM31	195.7	74	16(12/4)/39	159.0	68.7	266.6	27.7	26.0	7.6	R
Gongmushan Section										
GM1	192.7	29	8/10	262.5	42.9	267.5	15.2	93.0	5.8	R
GM2	213.7	30	10/10	264.2	43.5	274.6	18.4	91.1	5.1	R
GM3	224.7	40	10/10	263.8	40.1	277.5	10.5	16.0	12.4	R
GM4	224.7	26	10/10	275.3	30.4	281.0	9.3	8.5	17.6	R
GM5	248.7	20	6/10	292.8	29.7	298.5	15.0	28.5	12.8	R
GM6	194.7	44	5/10	280.5	53.0	282.1	9.0	24.1	15.9	R
GM7	225.7	37	10/10	266.1	36.7	277.5	9.5	43.4	7.4	R
GM8	225.7	37	10/10	274.0	48.1	288.0	17.1	30.9	8.8	R
GM9	229.7	35	5/10	261.8	36.6	275.2	14.1	43.2	11.8	R
GM10	200.7	31	10/10	255.8	46.5	266.0	19.3	38.4	7.9	R
GM11	223.7	34	10/10	259.3	48.1	277.4	23.6	18.2	11.7	R
GM12	221.7	37	7/8	268.2	46.3	282.1	16.0	35.1	10.3	R

[a]Site number with asterisk indicates the site mean direction is combined by two adjacent sites. N and n are numbers of specimens measured in each sites and specimens used for statistics, respectively; g/d is the number of specimens whose directions were obtained by great circle/the direct direction. Dec., declination; Inc., inclination; k, precision parameter; α_{95}, radius of the 95% confidence circle. R and N are reversed and normal polarity. D_g, I_g, D_s, and I_s are the declination and inclination in in situ coordinates and stratigraphic coordinates, respectively. For LM1–LM10, mean $D_g = -0.3$, $I_g = -32.2$, $k = 26.7$, $\alpha_{95} = 10.9$, $D_s = 44.0$, $I_s = -18.1$, $k = 26.7$, and $\alpha_{95} = 10.9$. For LM11–LM31, mean $D_g = 185.2$, $I_g = 54.3$, $k = 23.6$, $\alpha_{95} = 7.1$, $D_s = 250.1$, $I_s = 23.7$, $k = 27.2$, and $\alpha_{95} = 6.5$. For GM1–GM12, mean $D_g = 268.9$, $I_g = 42.3$, $k = 57.3$, $\alpha_{95} = 5.8$, $D_s = 279.2$, $I_s = 14.9$, $k = 74.2$, and $\alpha_{95} = 5.1$.

in the lower part of the section, and a large gap about 230 m, and then a reversed subchron zone of 970 m in the upper part of the section. Two short magnetic excursions are found in the normal and reversal subchrons. The orthogonal vector diagrams of representative samples of normal and reversal polarity at different stratigraphic levels (Figure 9) show that the excursions covered only about 10.3 and 19.4 m in the lower and upper basalt lava, respectively.

[21] It is clear that the HTCs direct to SW-W from both the LM and GM sections, which is not fully antipodal to the northeasterly upward directions from the lower part of section LM. A similar situation is also found at Duge and Ebian sections

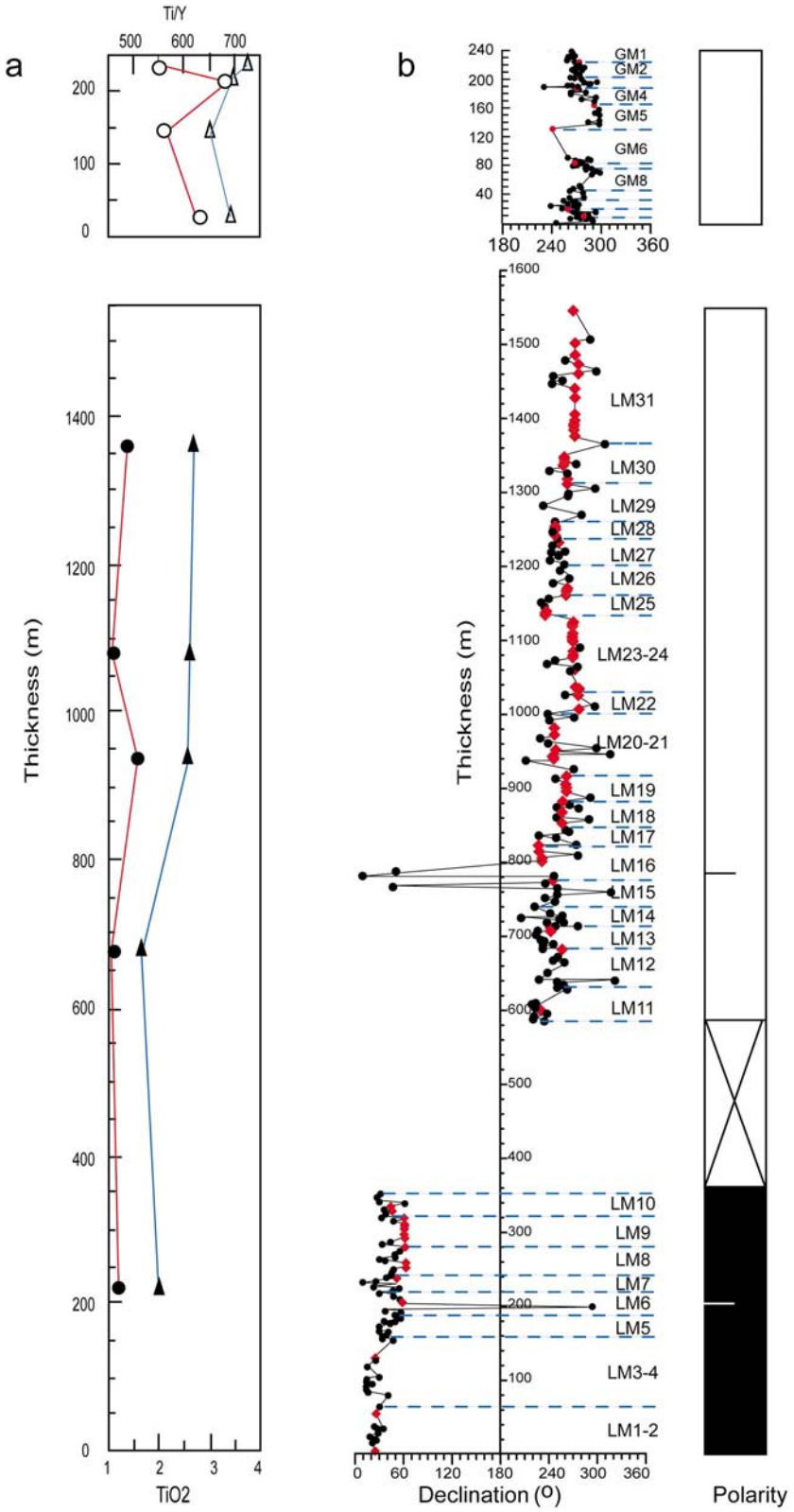


Figure 8. (a) Vertical variation of TiO_2 contents (triangles) and Ti/Y ratios (circles) of the Emeishan basalts in the sampling sections of Yanyuan. The open circles (triangles) are the data of GM, and the filled circles (triangles) are the data of LM. (b) Declination variation along the sampling sections and polarity definition. The sampling site numbers correspond to sites in Table 2. Declination direction defined by great circles is indicated by a red solid diamond.

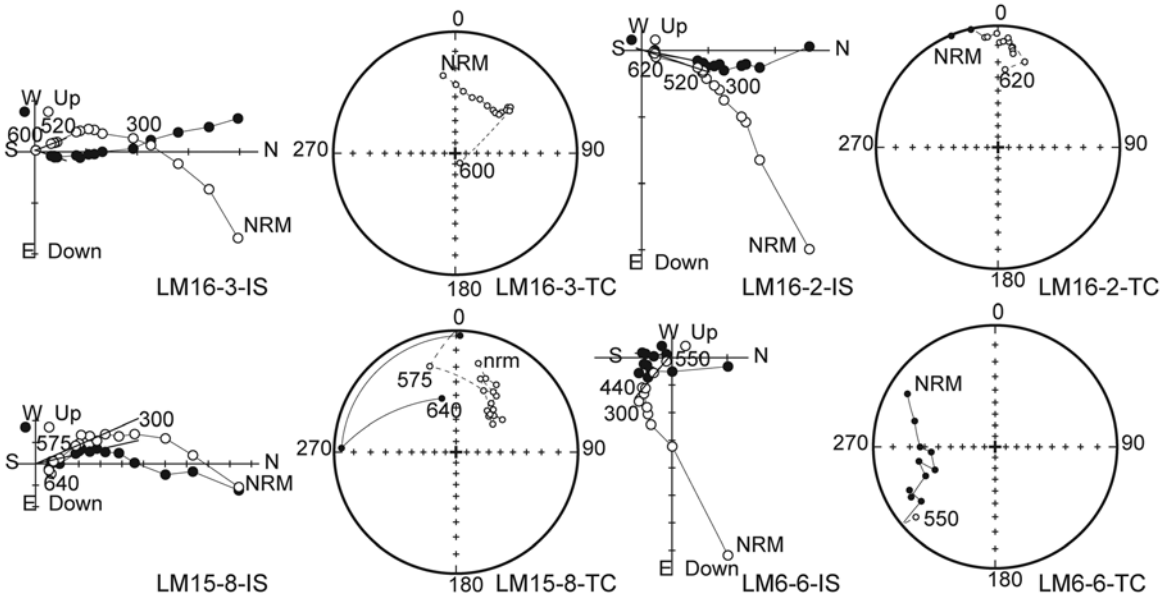


Figure 9. Representative orthogonal plots of thermal demagnetization results [*Zijderveld*, 1967] show the short magnetic excursions from Yanyuan basalts in in situ coordinates and stereoplots in stratigraphic coordinates. Symbols as in Figure 7.

[*Huang and Opdyke*, 1998; *Ali et al.*, 2002]. The difference in declinations of ChRM in the different places of the ELIP might have been caused by local rotation, which will need further study. However, all these results suggest that the NE upward direction was normal, and that the SWW downward direction was reversed polarity.

5. Discussion

5.1. Chemostratigraphic Correlation

[22] The TiO_2 contents of the Emeishan basalts are usually classified into HT and LT associations, and these display several striking features in the inner, intermediate and outer zones [*Xiao et al.*, 2003b; *He et al.*, 2007]. Systematic geochemical data were reported from Binchuan [*Xiao et al.*, 2003a] and Pingchuan (or Yanyuan area) both in the inner zone [*Xu et al.*, 2004; this study], Duge in the intermediate zone [*J. F. Xu et al.*, 2007; *Xu et al.*, 2003], and Zhijin in the outer zone [*Xu et al.*, 2003], which were used for a discussion of spatial and temporal variation, and chemostratigraphic correlation [*Xu et al.*, 2004; *Xiao et al.*, 2003a, 2003b]. These authors argued that LT basalts in the inner zone were emplaced in the early stage, and the overlying HT basalts in the inner zone, or unitary HT in the intermediate zone and outer zone represent the waning activity of the ELIP (Figure 10).

[23] The thickness of the lavas at Binchuan in Yunnan province is 5384 m, which is regarded as the eruption center [*Zhang et al.*, 1988; *He et al.*, 2007]. The basalt sequences at Binchuan are the most complete, which allows them to be considered as a reference for geochemical study of the Emeishan basalts. The TiO_2 contents and the Mg # ratios in the basalts at Yanyuan are similar to the values observed from the Binchuan samples. The LM section can be compared with the LT in the lower part of the sequences at Binchuan, the latter having the TiO_2 contents and ranges in Mg # between 1.53% and 1.7% and 0.48–0.54, respectively [*Xiao et al.*, 2003a]. The GM section is correlated to the HT basalts (TiO_2, 4.25%–4.63%; and Mg #, 0.39–0.53) [*Xiao et al.*, 2003a]. However, the differences between the LT and HT basalts at Yanyuan are insignificant. The trace elements including REE in the LM section are in the same range as those observed on LT2 basalts at Binchuan; similarities to the LT1 basalts of Binchuan area could not be found. The LT and HT samples at Yanyuan exhibit trace element ratios that overlap with the field of OIB (Figure 3), except for the negative Sr anomalies and positive Nb anomalies.

5.2. Magnetic Polarity Sequence and Regional Correlation

[24] Many paleomagnetic studies have been carried out on the Emeishan basalts in southwest China

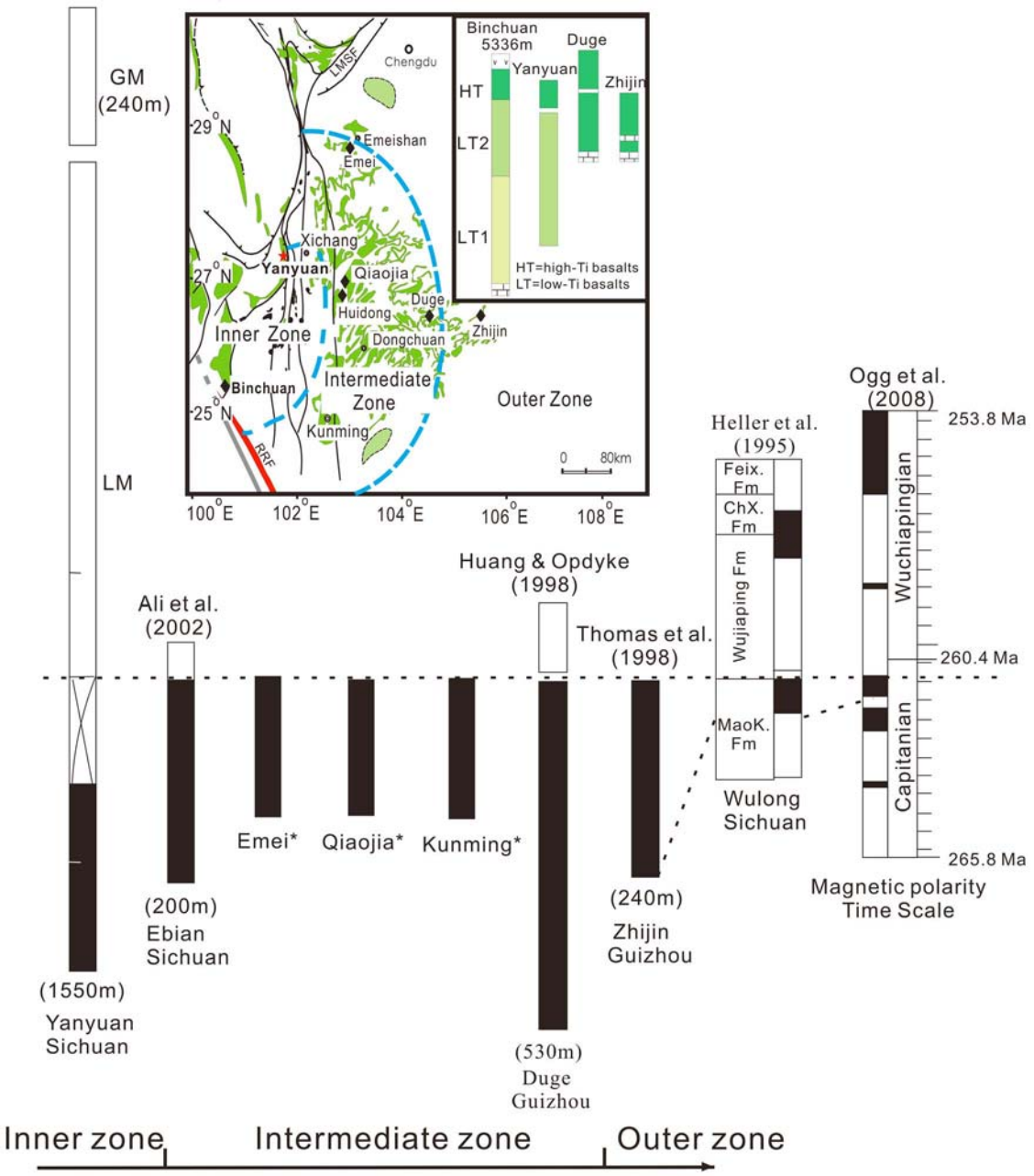


Figure 10. Correlation of magnetostratigraphic results obtained from different sections in the ELIP, Luomapu (LM) and Gongmushan (GM) sections (this study), Zhijin [*Thomas et al.*, 1998] and Duge sections [*Huang and Opdyke*, 1998] in Guizhou province, and Ebian section in Sichuan province [*Ali et al.*, 2002]. Polarity indications deduced from paleomagnetic studies at Emeishan [*McElhinny et al.*, 1981; *Zhao and Coe*, 1987; *Liu et al.*, 1987], Kunming [*Van der Voo et al.*, 1993], and Qiaojia areas [*Liu and Zhu*, 2009] are also shown (see also text). Black indicates normal polarity; white indicates reversed polarity. The Middle-Late Permian magnetobiostratigraphic results from Wulong in Sichuan [*Heller et al.*, 1995; *Jin et al.*, 1998; *Mei et al.*, 1998] and the magnetic polarity time scale [*Ogg et al.*, 2008] are shown for comparison. ChX. Fm, Changxing Formation; Feix. Fm, Feixianguan Formation; MaoK. Fm., Maokou Formation. Inset illustrates the distribution of LT basalts only in the inner zone and HT basalts in both the inner and outer zone (modified from *Song et al.* [2008], copyright 2008, with permission from Elsevier). The localities of paleomagnetic studies are indicated by diamonds. Stratigraphic variations of TiO_2 contents and Ti/Y of four representative lava successions are illustrated. Data sources are as follows: Binchuan, *Xiao et al.* [2004]; Yanyuan, *Y. G. Xu et al.* [2007] and this study; Duge and Zhijin, *Y. G. Xu et al.* [2007].

Table 3. Paleomagnetic Results From the Emeishan Basalts in Southwest China[a]

Localities	Coordinates	D_s (deg)	I_s (deg)	N	n	α_{95}	Paleopole	A_{95}	P	Reference
Emei	29.6°N, 103.4°E	16.4	−4.5	24		3.5°	54.5°N, 254.3°E	3.9°	N	*McElhinny et al.* [1981]
Emei	29.6°N, 103.4°E	19.6	−11.6		44	3.6°	49.7°N, 252.0°E	2.8°	N	*Zhao and Coe* [1987]
Ereshan	29.7°N, 103.5°E	38.4	−11.8		46	5.1°	38.7°N, 231.1°E	3.7°	N	*Liu et al.* [1987]
Huidong	26.7°N, 102.9°E	23.0	−4.5		7	10.0°	53.3°N, 241.8°E	8.5°	N, R	*Huang et al.* [1986]
Kunming	25.0°N, 102.7°E	25.5	−11.9	10	57	6.1°	50.3°N, 241.0°E	4.5°	N	*Van der Voo et al.* [1993]
Kunming	25.0°N, 102.7°E	244.1	2.6	1	2	-	22.7°N, 205.8°E	-	R	*Van der Voo et al.* [1993]
Zhijin	26.5°N, 105.7°E	46.2	−1.8	4	26	29.6°	37.8°N, 219.8°E	20.9°	N	*Thomas et al.* [1998]
Duge	26.4°N, 104.7°E	39.8	0.6	19	122	2.9°	43.7°N, 222.5°E	2.0°	N	*Huang and Opdyke* [1998]
Duge	26.4°N, 104.7°E	232.3	21.6	3	18	19.1°	26.6°N, 224.5°E	13.4°	R	*Huang and Opdyke* [1998]
Ebian	29.3°N, 103.3°E	16.4	7.7	17	93	6.1°	60.2°N, 248.8°E	4.4°	N	*Ali et al.* [2002]
Qiaojia	27.07°N, 102.98°E	357.9	−10.6	9	116	7.4°	57.4°N, 286.7°E	6.0°	N	*Liu and Zhu* [2009]
Luomapu	27.7°N, 101.9°E	44.0	−18.1	8	82	10.9°	33.6°N, 226.5°E	8.2°	N	this study
Luomapu	27.7°N, 101.9°E	250.1	23.7	19	168	6.5°	11.2°N, 212.5°E	5.1°	R	this study
Gongmushan	27.4°N, 101.8°E	279.3	14.9	12	101	5.1°	−11.7°N, 194.3°E	3.7°	R	this study

[a]N and n are numbers of sites and specimens, respectively. D_s and I_s are declination and inclination in stratigraphic coordinates. P is polarity, where R and N are reversed and normal polarity. α_{95} and A_{95} are radius of the 95% confidence circle of site mean and polar error, respectively.

(Table 3) with the main result that two ChRM directions have been found. One is a dominant northeasterly direction with shallow upward inclination [*Enkin et al.*, 1992]; the other shows a southwesterly direction with shallow inclination and was found also by *Huang* [1988], *Huang and Opdyke* [1998], *Van der Voo et al.* [1993], *Lin* [1984], and *Zhou et al.* [1988], but on a smaller set of samples. The ChRM directions pass both fold tests [*McElhinny et al.*, 1981], and/or the reversal test [*Huang et al.*, 1986; *Zhou et al.*, 1988] or conglomerate test [*Huang and Opdyke*, 1995], all indicating a primary thermal remanent magnetization. It is notable that the reversed polarity directions are usually not antipodal to the normal magnetized directions. The declination differences between the reversed reversal and normal directions are 10.8° at Huidong (with only 2 reversed samples), 12.5° at Duge, and 38.6° in Kunming (only with 2 reversed samples) [*Van der Voo et al.*, 1993], and 25° in the LM section. The reason for the failed antipodal directions recorded between the lower and upper part of the sections or different regions needs further study. The short geomagnetic spot record of normal polarity during the Permian Illawarra period [*Heller et al.*, 1995] might have induced the declination deviation, which cannot be fully excluded.

[25] Strictly speaking, there are only two magnetostratigraphic studies that have been carried out on the Emeishan basalts [*Huang and Opdyke*, 1998; *Ali et al.*, 2002]. Our results are consistent with these studies, which show that the lower part of the Emeishan basalts carried a predominantly normal polarity, with a reversed polarity zone in the upper part (Figure 10). In fact, the basalts at Duge that record both normal and reversal polarity zones are all HT basalts [*Y. G. Xu et al.*, 2007], which were considered to be the product of deep mantle material. However, the magnetic polarity stratigraphic correlation between Duge and Yanyuan sections indicates that the eruption of HT basalts of the whole section at Duge might cover both the early stage with normal polarity and later stages with a short reversal polarity. Specifically, the LT basalts at Yanyuan might have erupted at the same time as the HT basalts in the lower part of Duge section (Figure 10), because both record normal polarity. The temporal correlations between the LT and HT basalts in the inner and outer parts within the ELIP are contrary to the geochemical results, the latter indicating that the compositional variations in the lavas from the center to the margin of the ELIP clearly imply a delayed eruption process [*Xu et al.*, 2004]. Our results suggest that the basalts in the inner zone composed of LT basalts in the lower part may be correlated to most of the HT basalts in the intermediate and outer zones of the ELIP. The LT and HT basalts with reversed polarity in the inner zone may be correlated to a few HT basalts in the intermediate zone (e.g., Duge area), or were completely absent or eroded in the intermediate and outer zones of the ELIP (Figure 10).

[26] It is noted that available Permian paleomagnetic results from South China are mostly from the Emeishan Basalts (Table 3). These results used for tectonic study are potentially indicative of polarity determinations. Results from the high content of Ti lava sections in the intermediate-outer zones, such as Emeishan [*McElhinny et al.*, 1981; *Zhao and Coe*, 1987; *Liu et al.*, 1987], Zhijin [*Thomas et al.*, 1998], Qiaojia [*Liu and Zhu*, 2009], and Kunming areas [*Van der Voo et al.*, 1993], all indicate normal polarity directions (Table 3). The magnetic polarity correlations (Figure 10) show that these Emeishan

basalts have to be correlated to the low content of the Ti basalts in the lower part of the Yanyuan section, which were erupted at the same time with the same normal polarity in the early stage of ELIP formation.

[27] Thus, the magnetic polarity sequences from the ELIP indicate clearly that the eruption of Emeishan basalts can be divided into early and late stages. The early eruption included both the LT basalts in the inner zone and most of the HT basalts in the intermediate and outer zones, which all record a normal polarity in the early stage, whereas the late erupted LT and HT basalts in the upper parts of the sections in the inner zone and the few in the intermediate zone record a reversal polarity in the ELIP.

[28] Any model dealing with the formation and thermal structure of the ELIP and the nature of mantle-lithosphere interaction should consider the temporal and spatial relationship between the LT and HT basalts. The rifting or plume-related origin generation of the ELIP has been debated, and spatial variation of compositions may be related to heterogeneous mantle sources, the thickness of the lithosphere beneath the southwestern margin of the Yangtze block, mantle plume-lithosphere interaction and crustal contamination, or previous subduction slab along the continental margin [*Song et al.*, 2001a, 2001b, 2008; *Xu et al.*, 2001; *Xiao et al.*, 2003b; *Wang et al.*, 2007; *Qi and Zhou*, 2008].

[29] Contradictory results from the geochemical studies have been reported. However, most agreed on a deeper mantle plume source for the high-Ti basalts with a smaller degree of partial melting from the mantle than the LT lavas, and more extensive lithospheric mantle and crustal contamination in the low-Ti basalts than in the high-Ti basalts [*Xu et al.*, 2001; *Xiao et al.*, 2004]. Nonetheless, combining Pb, Nd and Os isotopic data, *J. F. Xu et al.* [2007] demonstrate that the low-Ti basaltic magmas in the ELIP were mainly sourced from a mantle plume reservoir, whereas the high-Ti basaltic magmas were most likely derived from a subcontinental lithospheric mantle (SCLM) reservoir or were contaminated by a significant amount of lithospheric mantle material during plume-related magma ascent through the SCLM [*Qi and Zhou*, 2008].

[30] It should also be noted that the western margin of the Yangtze block was a passive margin, where the subduction of the Indochina block along the Jinshajiang-Ailaoshan-Song Ma zone was initiated in the Early Permian [*Wang et al.*, 2000]. The different lithospheric structures would be the reason for different spatial variation of LT and HT basalts within the ELIP, which might contribute different degrees of mantle plume-lithosphere interaction. The distribution of the LT basalts is generally along the western margin of the Yangtze block.

[31] Thus, we consider that the large volume of Emeishan basalts was erupted over a large extent in the first stage, with the LT basalts derived from a depleted mantle source [*Xu et al.*, 2001] and substantial crustal contamination in the oceanside margin of the western ELIP; at the same time, the HT basalts probably formed first at depths of 200 to 400 km during the initial upwelling by low degrees of partial melting of enriched mantle material erupted in the eastern part (continental side) of the ELIP. The second stage, lava sequences of LT and HT volcanic rocks occurred in the upper part of the sections in the western ELIP (inner zone), and a few high-Ti basalts in the east of the ELIP. The depleted mantle source might have been locally modified by subducted oceanic crust so as to generate an enriched mantle source from which minor LREE-enriched picrites formed [*Wang et al.*, 2007; *Zhang et al.*, 2008].

5.3. Fast Eruption of the ELIP

[32] Significant progress has been achieved recently on the dating of the Emeishan basalts. Stratigraphic correlation and SHRIMP zircon U-Pb dating on the Upper Permian Xuanwei Formation that overlies directly the Emeishan basalts gave 257 ± 3 Ma for the lowermost formation [*He et al.*, 2007]. *He et al.* [2007] also reported an age of 260 ± 5 Ma for the clay tuff at the Middle-Late Permian boundary at Chaotian in the northern Sichuan basin. Although still uncertain, these results indicate that the Emeishan basalts were probably emplaced close to the Middle-Late Permian boundary at 260 Ma [*Zhou et al.*, 2002; *Jin et al.*, 2006; *He et al.*, 2007; *Ogg et al.*, 2008]. However, the eruptive duration of the ELIP remains undetermined.

[33] A magnetostratigraphic study was carried out in the Permian Maokou Formation in the Wulong area of Sichuan, south China [*Heller et al.*, 1995]; the end of this formation is correlated to the end-Guadalupian, or close to the boundary of the Capitanian/Wuchiapingian [*Jin et al.*, 2006; *He et al.*, 2007]. Comparison of magnetostratigraphic results between Yanyuan and Wulong reveals directly that the normal polarity zone in the early time of eruption at Yanyuan can be correlated to the normal polarity zone in the Maokou Formation, and the reversed zone can very probably be correlated to

the reversed zone in the Wujiaping Formation (Figure 10). However, comparing with the Permian magnetic polarity time scale [*Ogg et al.*, 2008], our result shows that the normal and reversal polarity sequences of the Yanyuan section can be matched with the normal polarity just below the Capitanian-Wuchiapingian boundary and is followed by reversal polarity upward to the boundary (Figure 10). Because the normal polarity under the Capitanian-Wuchiapingian boundary is estimated as less than 0.5 Ma [*Ogg et al.*, 2008], the magnetic correlation also indicates that the eruptive duration of the ELIP can be thus estimated as occurring within a short time, probably less than 1 Myr (Figure 10). This proposition is supported by the suggestion that the initial eruption of the ELIP might have started just before the *Jinogondolella altudaensis* conodont zone in the mid-Capitanian, as also suggested by *Wignall et al.* [2009]. However, the duration of the ELIP might have covered the normal and reversal polarities period, with the eruption ended by the end-Guadalupian.

6. Conclusions

[34] Geochemical results indicate that the basalts in the LM section are composed of LT basalts (Ti/Y ~ 484.2–511.4) and the GM section is composed of high-Ti basalts (Ti/Y ~ 558.2–658.9) at Yanyuan, which show consistent characteristic distributions in comparison with those of the Binchuan areas in the south within the inner zone of the ELIP. Paleomagnetic studies reveal two distinct characteristic remanent magnetizations with northeasterly upward direction and SWW downward direction. These results suggest that the basalts carried a normal polarity in the lower part of the section, and a reversed polarity zone in the upper part.

[35] Combined with available magnetostratigraphic and paleomagnetic results carried out in the ELIP, all those results show a characteristic pattern of normal and/or reversal polarity. Thus, the magnetic polarity sequences from the ELIP indicate clearly that the eruption of the Emeishan basalts can be divided into early (normal polarity subchron) and late (reversal subchron) stages. The early stage of eruption may represent the main phase of the volcanism, including the lower part of the LT basalt sequence in the inner zone (west province) and most of the HT basalts in the intermediate and outer zones (east province). The late eruption contained partly low-Ti and overlain high-Ti basalts in the upper sequence of the sections in the inner zone and few high-Ti volcanic successions in the intermediate zone. This interpretation is significantly different from the proposition that the variation from LT to HT lavas might be related to the transition from the peak to the waning stage of mantle plume activity [*Xu et al.*, 2001; *Xiao et al.*, 2004]. Temporal and spatial variations of the LT and HT basalts in the ELIP are crucial for the modeling of ELIP formation.

[36] By comparing these data with Middle-Late Permian magnetic stratigraphy, our magnetic polarity sequence results also indicate that the Emeishan large igneous province was formed within a very short time, probably less than 1 Myr. However, precise age determinations are still needed for further confirmation.

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